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Microtextured Substrates and Microparticles Used as in Situ Lenses for On-Chip Immunofluorescence Amplification

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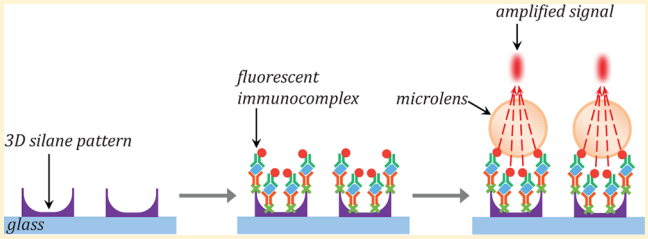
ABSTRACT: We propose a concept for enhancing the fluorescent detection signal of on-chip immunoassays by using three-dimensional 3 and 8 μm size microtextured structures as substrates for the assay, which allows exploitation of a large number of fluorophores within the focal plane of an optical microscope for detection. Additionally, we demonstrate the use of 3 and 9.75 μm dielectric microparticles, respectively, as in situ lenses on the 3 and 8 μm microstructures, respectively, for further enhancement of the optical signal. In our model system, the fluorescent complex is formed on (3-aminopropyl)triethoxysilane microstructures and we use carboxyl-functionalized melamine microparticles as the in situ lenses. Mouse IgG diluted in phosphate-buffered saline is used as model target antigen and can be easily detected down to a concentration of 2 ng/mL thanks to these approaches. Also, we present a detailed two-dimensional numerical study of the light propagation through a dielectric microparticle using the finite element method, providing key insight into the signal amplification mechanism of a microlens, and point out its advantageous use in microfluidic assays.



Microfluidic immunoassays are playing an increasingly important role in healthcare diagnosis, drug toxicity evaluation, food quality testing, and environmental safety monitoring.[1−5] Microfluidic channels are characterized by a large surface-to-volume ratio, which makes them particularly interesting to be used in heterogeneous assays. Indeed, a properly functionalized microchannel wall can easily purify a specific target antigen from a complex sample matrix. Among a wide variety of bioanalytical immunoassay principles, including the enzyme-linked immunosorbent assay (ELISA) as a gold standard for protein analysis, fluorescent detection methods have become predominant due to their high sensitivity and ease of operation.[6,7] However, when the fluorescence is quantified using an optical microscopy system, the presence of fluorophores in only a thin surface layer on the microchannel wall limits the detection signal, despite the bright intensity of the fluorophores, essentially because of the number of fluorescent complexes present in the microscope focal plane is limited by the two-dimensional (2D) format.[8−11]

Also the use of microparticles in microfluidic immunoassays has attracted great interest recently.[12−16] Microparticles offer an increased reaction surface in an assay; they can be used to concentrate or purify a target antigen on their surface and thereby allow reduction of sample and reagent volumes. Moreover, a wide variety of microparticles with different surface functional groups against a large number of target molecules exist.[17−22] However, when microparticles are randomly suspended inside a microfluidic channel, this can pose problems in their handling or quantification of their number. Here, microparticles with magnetic properties can provide a solution, as they provide an extra handle for performing magnetic force-based procedures for microparticle manipulation.[12,19,23] On the other hand, surface-immobilized microparticles have been used for microfluidic heterogeneous immunoassays, either as substrates or as labels.[24−28] Before, we have reported a technique based on electrostatic assembly of streptavidin-functionalized microparticles on silane patterns[26] and showed that, using these microparticles as substrates, it was possible to perform a sandwich immunoassay, in which the fluorescent signal from a secondary antibody (Ab) quantified the target antigen (Ag) of interest. Alternatively, the binding of functionalized microparticles to surface-immobilized targets within microchannels has enabled the quantification of biological targets based on counting or detecting the number of captured microparticles.[29−31] Also, dielectric (nonmagnetic) microparticles with low optical absorbance and high refractive index have been used as in situ lenses for improved photon detection.[32] More commonly, microfabricated lenses have been used in microfluidic chips and demonstrated an enhanced detection efficiency of weak optical signals.[33−36] On the other hand, the commercial availability of optically transparent microspheres with a variety of surface functionalizations makes these particularly attractive candidates too for in situ optical signal amplification.[32,37]

We propose here two measures for enhancing the fluorescent detection signal in microfluidic immunoassays: the use of (i) three-dimensional (3D) microstructures on a glass wafer as substrates for the assay, which allow detecting the fluorescence of a large number of fluorophores within the focal detection volume of an optical microscope, and (ii) dielectric micro-

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particles as in situ lenses for further concentration of the optical signal. We use mouse IgG diluted in phosphate-buffered saline as model target antigen in the immunoassay. Moreover, a detailed 2D numerical study of the light propagation through a dielectric microparticle using the finite element method (FEM) is presented, highlighting the mechanism of optical signal amplification.

■ MATERIALS AND METHODS

Materials. (3-Aminopropyl)triethoxysilane (APTES) solution, phosphate-buffered saline (PBS) 10× concentrate solution, nonionic surfactant Tween-20, bovine serum albumin (BSA), and chlorotrimethylsilane (TMCS) were purchased from Sigma-Aldrich Chemie GmbH (Buchs, Switzerland). The 1× PBS solution was diluted from the 10× PBS concentrate solution; PBS−Tween (PBST) solution was prepared by mixing 0.5% v/v Tween-20 with 1× PBS, and PBS−BSA solution was prepared by diluting 1% w/v BSA in 1× PBS. Streptavidin was purchased from ReactoLab SA (Servion, Switzerland). Biotinylated goat antimouse immunoglobulin (IgG, capture Ab), normal mouse IgG (target Ag), and Alexa Fluor 647-labeled goat antimouse IgG (detection Ab) were purchased from Invitrogen AG (Basel, Switzerland). Carboxyl-functionalized microparticles based on melamine resin with two sizes, i.e., 3 and 9.75 μm, were purchased from Sigma-Aldrich Chemie GmbH and microParticles GmbH (Berlin, Germany), respectively. Poly(dimethylsiloxane) (PDMS) prepolymer and curing agent were purchased from Dow Corning (Wiesbaden, Germany). Float glass wafers (Ø4 in.), photoresist (PR) AZ ECI 3027, and deionized (DI) water were obtained from EPFL's Center of MicroNanoTechnology.

3D APTES Microstructures and Microfluidic Setup. Photolithography processes are used to pattern the 3D APTES microstructures on a glass wafer. Two different formats of microstructures are fabricated, namely, 3 μm wide dots and 8 μm wide squares. First, a float glass wafer is spin-coated with PR at 5500 rpm for 30 s to achieve a 0.62 μm thick film. Photoresist patterns of dots and squares are generated by standard photolithography, after which the glass wafer is diced into small chips (16 mm × 30 mm) and ready to be used for 3D APTES microstructures fabrication. In this process, the glass chip is subjected to air plasma at 10 W for 60 s to hydroxylate the exposed glass surface. Following this, 10% (v/v) APTES solution in DI water is spin-coated at 3000 rpm for 30 s (Figure 1a) onto the substrate, and it is baked at 100 °C on a hot plate for 2 min for condensation (Figure 1b). As the final step, the PR and unreacted APTES are ultrasonically removed in acetone, resulting in 3D APTES micropatterns on the glass chip.

A microfluidic channel is realized on the glass chip by reversibly bonding a PDMS half-channel to the latter. The half-channel is formed starting from an SU-8 structure that is fabricated using soft-lithography on a silicon wafer and used as the mold for the PDMS replication. After silanizing the SU-8 master using TMCS in a vacuum desiccator, a 10:1 mixture of PDMS prepolymer and curing agent is cast over the master and cured at 70 °C for 4 h. Then the PDMS replica is peeled off from the mold, resulting in a microfluidic half-channel with a height of 50 μm, a width of 100 μm, and a length of 24 mm. Access holes for inlet and outlet connections are made by a Harris Uni-Core (Pelco International, Redding, U.S.A.) punching tool. Two poly(methyl methacrylate) (PMMA) plates are used to clamp the PDMS replica and the glass substrate; the alignment between the PDMS half-channel and the 3D APTES microstructures is not critical, as the latter is present on an area with a length of 2400 μm and a width of 800 μm on the center of the glass substrate. Using this method a leak-proof seal is obtained. A neMESYS syringe pump (Cetoni GmbH, Korbussen, Germany) generates the positive pressure used for injecting the reagent and microparticles solutions through the microchannel. All the filling and washing steps in the immunocomplex experiments are performed at a flow rate of 1 nL/s.

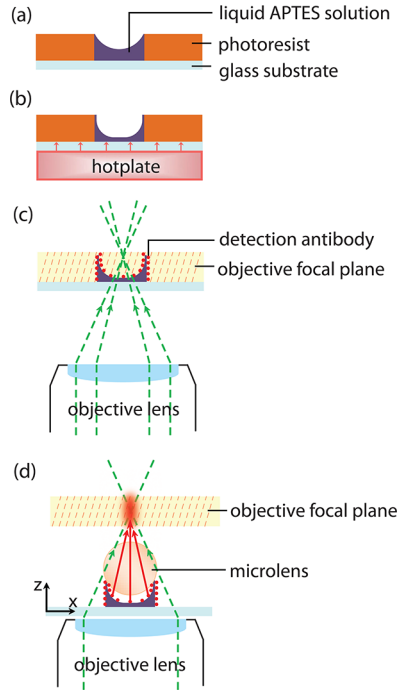

Figure 1. Schematic illustration of the use of microtextured substrates and in situ microlenses for immunofluorescence amplification. (a) Liquid APTES solution is spin-coated onto a glass substrate having photoresist micropatterns. (b) During the hard-bake process, solvent evaporates, while the dissolved APTES molecules accumulate to the photoresist side walls by capillarity, after which the photoresist is ultrasonically removed in acetone, resulting in 3D APTES microstructures. (c) A 3D APTES microstructure is used as substrate for immunocomplex formation, and a microscope is used to record the fluorescence intensity; in this case, the focal plane of the objective is aligned with the position of the APTES layer. (d) A carboxyl-functionalized melamine microparticle is self-assembled on top of the APTES microstructure and used as microlens to enhance the fluorescent signal from the fluorophores; in this case, the objective focal plane coincides with the focal plane of the microlens.

Immunoassay Protocol and in Situ Microlens Patterning. The immunoassay experiments are performed in a stop-flow mode: the microfluidic channel (volume of 120 nL) is filled and incubated with a biological sample solution subsequently, and the sandwich immunocomplex is formed on the 3D APTES microstructures. As the first step of the assay, 100 μg/mL of streptavidin in PBS is flowed into the microfluidic channel and incubates for 5 min to electrostatically bind with the amine groups on the APTES microstructure. The streptavidin serves as a linker molecule between the APTES microstructure and the capture Ab in the immunoassay. After briefly washing with PBST, the microchannel is blocked using PBS−BSA for 2 min to avoid the unspecific adsorption of Abs/Ags on the microchannel walls. Following this, biotinylated goat

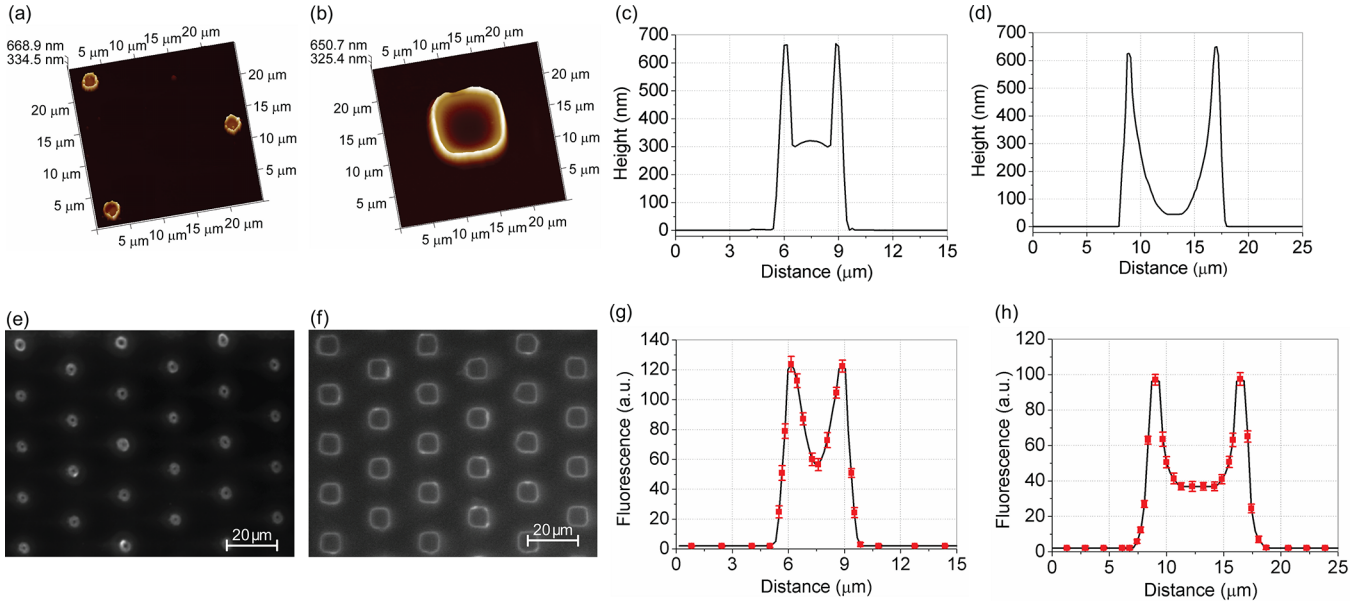

Figure 2. (a and b) AFM images showing the topography of 3 μm dot and 8 μm square APTES microstructures, respectively, in particular the ridges that are formed due to the presence of the photoresist side walls in the lift-off process. (c and d) AFM-determined profiles in the center of a dot and square APTES microstructure, respectively. (e and f) Fluorescence microscope images after performing the immunoassay on dot and square APTES microstructures, respectively; 10 ng/mL mouse IgG in PBS is applied into the microchannel as target Ag. (g and h) Fluorescence intensities of dot and square microstructures which are positioned on the six most upstream columns, respectively. Points represent the average and error bars the variance, while full lines are a guide to the eye.

antimouse IgG is diluted in PBS and used as capture Ab due to its high affinity to streptavidin. An amount of 10 μg/mL of capture Ab solution is introduced into the channel and incubates for 5 min. Unbound capture Ab is removed by washing with PBST buffer, after which the microchannel is filled and incubated with mouse IgG diluted in PBS, which serves as the model target Ag solution. We perform the assay using 360 nL of Ag solution with different concentrations, and use three consecutive fill/incubation steps, each taking 2 min for filling and 10 min for incubation. Finally, after washing with PBST, 10 μg/mL Alexa Fluor 647-labeled goat antimouse IgG in PBS is pumped into the channel and incubated for 5 min. Unbound detection Ab is removed by washing with PBST. After the incubation with detection Ab, the fluorescence on the 3D APTES microstructures along the microchannel is recorded by a microscopic setup; the focal plane of the microscope is aligned to the APTES layer, as shown in Figure 1c. Then the 3 and 9.75 μm carboxyl-functionalized microparticles based on melamine resin, diluted 40 times from stock solution, are loaded into the microfluidic channel and immobilized on the 3D APTES microstructures. The objective focal plane of the microscope is moved to the focal plane of the microparticles, and the fluorescence intensity is recorded (see Figure 1d). The image acquisition and fluorescence detection of the sandwich immunoassay are achieved using a CCD camera ORCA-C4742−80ER (Hamamatsu Photonics Germany GmbH, Herrsching am Ammersee, Germany) mounted on an inverted microscope Axiovert S100 (Carl Zeiss GmbH, Oberkochen, Germany), equipped with a 20× objective (Achroplan, NA 0.45). The microscope is configured with a Mercury vapor short arc lamp X-Cite 120 (Carl Zeiss GmbH, Oberkochen, Germany) and filter set LC-XF407 (Laser Components, Olching, Germany) for Alexa Fluor 647. The exposure time used for capturing the fluorescence signal is 1.5 s, and Hokawo software (Hamamatsu Photonics Germany GmbH, Herrsching am Ammersee, Germany) is used for the fluorescence image analysis. The fluorescence intensity for a given Ag concentration is obtained by subtracting the background signal from nonspecifically adsorbed detection Abs on the glass.

■ RESULTS AND DISCUSSION

3D APTES Microstructures. In our design, the APTES microstructures, either dots with a diameter of 3 μm or squares with a width of 8 μm, are separated in the x- and y-direction by an interspacing of 20 μm. Atomic force microscopy (AFM) is performed at room temperature using a Bruker FastScan AFM (Bruker AXS Inc., Madison, U.S.A.) to determine the topography of the APTES microstructures, as shown in Figure 2, parts a and b; the corresponding AFM cross-sectional profiles indicating the "ridge effect" are shown in Figure 2, parts c and d. The APTES thickness on the ridges is ∼600 nm, which is approximately the thickness of the PR, while the thickness decreases toward the center of the microstructures. The origin for the ridge effect is the capillary force between the applied APTES solution and the vertical side wall of the PR. During the condensation process (see Figure 1, parts a and b), the APTES molecules accumulate toward the PR side walls while the solvent, i.e., water, evaporates. The 3D topography is size-related: at the center of the 3 μm dots, the thickness of APTES is ∼300 nm; when the size of the micropattern increases, the thickness at the center decreases; for the 8 μm squares, it is ∼50 nm.

Streptavidin is negatively charged above pH 5, while the APTES microstructure is positively charged at pH 7.4 due to the presence of protonated amine groups.[38] Thus, in the actually used pH 7.4 solutions, streptavidin and APTES bind electrostatically. Due to the high affinity between streptavidin and biotin, a biotinylated capture Ab can then be easily grafted on the dot or square microstructures, followed by subsequent application of target Ag and detection Ab solution in the

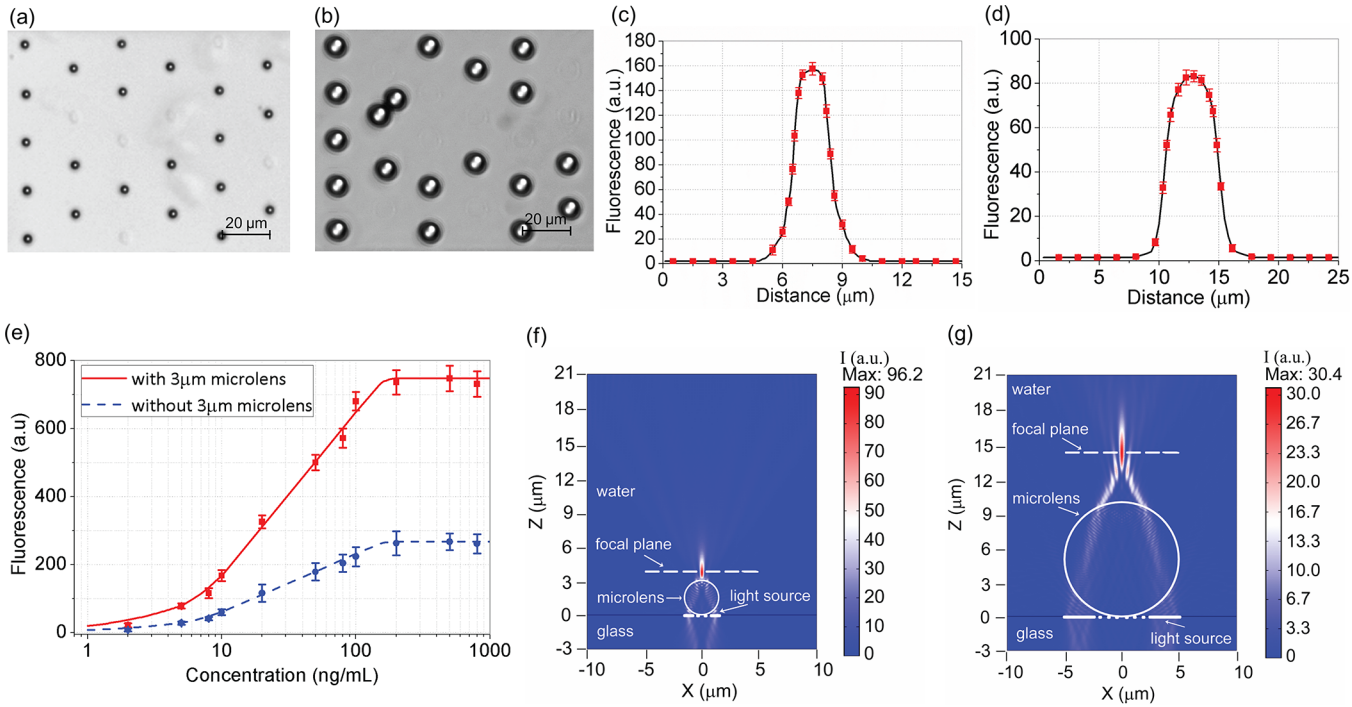

Figure 3. (a and b) Bright-field images of carboxyl-functionalized melamine microparticles that are self-assembled on 3 μm dot and 8 μm square APTES microstructures, respectively. (c and d) Fluorescence intensity at the focal plane of the microlenses, which are assembled on dot and square APTES microstructures. Points represent the average, error bars the variance, while full lines are guides to the eye. (e) Fluorescence intensity for different target Ag concentrations, obtained by focusing the microscope objective at the focal plane of the 3 μm microlenses that are situated on top of APTES dots and at the center of the APTES dots in absence of the microlenses; full lines are guides to the eye. (f and g) 2D FEM simulations (Comsol Multiphysics) of a 3 and 9.75 μm dielectric microparticle, respectively, used as a microlens for optical signal enhancement and positioned on a linear light source corresponding to the size of the used APTES microstructures.

microchannel. The ridge effect of the 3D APTES topography directly translates into variations of fluorescence intensities from the sandwich immunoassay, as shown in Figure 2, parts e and f, respectively. The fluorescence signal from the micropatterns which are positioned on the six most upstream columns (a column being defined as all dots or squares in the vertical (y) direction for a fixed value of x) is used to quantify the influence of the ridge effect on fluorescence distribution. Parts g and h of Figure 2 show the fluorescence intensity along the central line of the dot and square microstructures of Figure 2, parts e and f, respectively. Each point and error bar corresponds to the statistical average intensity and variance from pattern to pattern, respectively. An obvious increase of the signal at the ridge is observed as direct consequence of the 3D topography. An Ab monolayer typically has a height of 5−15 nm, depending on the Ab orientation (IgG has approximately a height of 14.5 nm, a width of 8.5 nm, and a thickness of 4.0 nm[39]). Thus, the ridge of the APTES microstructures of ∼600 nm is significantly higher than an IgG-based immunocomplex (capture Ab, target Ag, and detection Ab), resulting in a higher fluorescence intensity. Moreover, the observed fluorescent intensity at the center of 3 μm dots is larger than that found for the 8 μm square structures, probably due to the small size of a dot and its associated diffraction effect.

In Situ Assembly of the Microlenses. After completion of the immunoassay on the APTES microstructures, the carboxyl-functionalized melamine microparticles are introduced into the microfluidic channel. The melamine microparticles are negatively charged at pH 7.4 due to the partial ionization of the carboxylic acid surface groups. Thus, microparticles that are close to the APTES ridges are attracted electrostatically. However, also binding between the Abs situated on the APTES microstructures and carboxylic acid groups is possible. Nonspecific adsorption of microparticles can occur when the microparticles are incubated under stop-flow conditions, especially for long-time incubation. To minimize this effect, the delivery of 3 μm microparticles is performed under a continuous flow of 2 nL/s for 5 min, during which the microparticles are captured. Thereafter, a final washing step is performed for 5 min at a flow rate of 5 nL/s. A different experimental protocol is performed for 9.75 μm microparticles, which are intrinsically subjected to a higher drag force. The microparticles are loaded into the channel under continuous pumping for 2 min at 1 nL/s, then the flow is stopped and the microparticles incubation is done for 1 min; this procedure is repeated by three times. Note that we encountered difficulties when trying the in situ assembly of microparticles with a diameter of 20 μm on APTES microstructures of corresponding size. Indeed, the attachment based on electrostatic forces is hardly feasible, as the size of such microparticles becomes comparable to the half-height of the microfluidic channel, and the drag force acting upon these microparticles from the fluid motion becomes even bigger so that the electrostatic interaction proved insufficient to resist to even tiny transient flows in the microchannel.

Spherical aberration and comatic aberration can be observed when the microparticles are used as lenses. Especially when the microparticle is not centered with respect to the APTES microstructure, these effects are observed: they generate distortion of the image of the light source and affect the efficiency of the microlens in concentrating the fluorescence from the APTES microstructures. Therefore, a good alignment

of the microparticles with the APTES microstructures is of great importance. We were able to achieve this by controlling the experimental assay parameters carefully, especially the flow rate for microparticles introduction and during final washing. However, chromatic aberration can be neglected due to the narrow (~652−687 nm) fluorescence emission spectrum of the detection Ab. Moreover, by our choice for the size of the 3D APTES microstructures, the positioning of a single microparticle per pattern is typically achieved. The results of the in situ assembly of the two types of microlenses on the 3D APTES microstructures are shown in Figure 3, parts a and b.

Fluorescence Detection Using Self-Assembled Microlenses. The melamine microparticles have a refractive index n_p = 1.68, which means they can be used as microlenses in a water-based solution ($n_m \approx 1.33$). We note that the main role of the lens is focusing the light originating from its collection cone. The primary advantage of using a microlens is that this local enhancement of fluorescence can be simply monitored by using low magnification/low numerical aperture microscope objectives. As a consequence, many light sources can be observed at once with such microscope objective, focusing is easy, and statistical information on the intensity distribution over the different spots can be gained. The immunofluorescent signal on the focal plane of 3 and 9.75 μm microparticles which are positioned on the six most upstream columns are shown in Figure 3, parts c and d. Each point and error bar corresponds to the statistical average intensity and variance. It shows that, when a melamine microparticle is located on top of an APTES microstructure, the fluorescent signal, mainly generated from the ridge, can be collected and converged on the focal plane of the microlens (see Figure 1d). By comparing the fluorescent signal sampled at the focal plane of a 3 μm microlens (Figure 3c) with the signal originating from the center of a dot APTES microstructure in absence of the microlens (Figure 2g), we find that application of a 3 μm microlens allows amplifying the immunofluorescence by a factor ~2.8. On the other hand, the 9.75 μm microlenses show an amplification factor ~2, when we compare the fluorescence intensity at the center spots of Figures 3d and 2h. However, it should be noticed that the fluorescence intensity on the focal plane of the microlens of Figure 3d is smaller than that on the ridge of the APTES microstructures of Figure 2h. This implies that the 9.75 μm melamine microparticle cannot be used as an efficient lens to enhance the fluorescence signal. We believe this is due to two reasons: first, when the objective focal plane coincides with the focal plane of the 9.75 μm microlens, it is actually not well-focused on the fluorophores; hence, the excitation of the fluorescence is less efficient; second, the light absorbance in the dielectric medium (the melamine) is proportional to its thickness, which means the bigger microparticle absorbs more light and there is less signal enhancement.

To obtain the detection limit from this method, experiments at different target Ag concentrations are performed. The fluorescent detection curves in Figure 3e correspond to applying 360 nL of target Ag at different concentrations (0−1000 ng/mL) in PBS in three consecutive fill/incubation steps with the stop-flow mode. The 0 ng/mL concentration means that the experiment is performed without applying target Ag solution in PBS. In this case, the fluorescent signal corresponds to the nonspecific adsorption of fluorescent detection Abs on the APTES microstructures. Each point and error bar correspond to the statistical average and variance from three nominally identical immunoassay experiments. In Figure 3e, we compare the fluorescence signal sampled at the focal plane of a 3 μm microparticle and the signal obtained from the center of a 3 μm wide dot APTES microstructure in absence of the microlens, and an amplification factor of ~2.8 by the presence of the microlens is obtained. Mouse IgG in PBS is detectable down to a concentration of ~2 ng/mL.

Numerical Study of the Microlens Effect. The FEM is used to simulate the electromagnetic field distribution in the vicinity of a dielectric microsphere and provide insight into the mechanism of the microlens effect. In particular, we use the software COMSOL Multiphysics (version 4.2) to study the electromagnetic wave propagation through the water media (n_m = 1.33), the melamine microparticle (n_p = 1.68), and the glass substrate (n_s = 1.44). A 2D model is built to mimic the experimental situation, where a melamine microparticle is immobilized on a glass substrate and immersed in water. All the dimensional features in the model are kept the same as those in the experiment, except the APTES microstructure, which is simplified as a line source due to its small thickness compared to the diameter of the microparticle. Moreover, the line source is divided into three components, the central part and two sides, corresponding to the center and the ridges of the APTES micropatterns. A scalar equation is used to study transverse electric (TE) waves propagating through the three different media. Also, the light absorption in the microparticle is taken into account and the scalar equation for in-plane TE waves is used:

$$\nabla \times (\mu_r^{-1} \nabla \times \mathbf{E}) - (\varepsilon_r - j\sigma/\omega\varepsilon_0) k_0^2 \mathbf{E} = 0 \quad (1)$$

where μ_r is the relative permeability, ε_r is the relative permittivity, ε_0 is the vacuum permittivity, σ is the electric conductivity, ω is the angular frequency, and k_0 is the free-space wavenumber. In real materials, the polarization does not respond instantaneously to an applied field. This causes dielectric loss, which can be expressed by a permittivity that is both complex and frequency-dependent. Real materials are not perfect insulators either, i.e., they have nonzero direct current conductivity. Taking both aspects into consideration, a complex index of refraction can be defined and used to take into account light absorption in eq 1:

$$\varepsilon_r = (n - i\kappa)^2 \quad (2)$$

Here, κ is the extinction coefficient, which has a straightforward but nontrivial relationship with the absorption coefficient, which can be calculated directly from the absorbance of the dielectric medium. From the measurement of the absorption spectrum of the melamine microparticles, the extinction coefficient at a wavelength of 668 nm (the emission wavelength of Alexa Fluor 647) is 0.003.

In this model, the scattering boundary condition is used at all exterior boundaries and the continuity boundary condition is used at each material interface. The linear light source is divided into a central part with weak signal and the ridges with enhanced signal, which are shown in Figure 3, parts f and g, as the dotted and solid lines below the microlens, respectively. Their length and the intensity of the electric field correlate to the experimental fluorescent signal. A value of 668 nm is defined as the wavelength of the propagating wave. After meshing, the element size is one-third of the wavelength, which is sufficiently small to obtain a correct solution. After the model is solved, the distribution of the light intensity is obtained by multiplying the electric field by its complex conjugate, giving

the result of Figure 3, parts f and g. The simulation result shows the light focusing through the dielectric microparticle about 1 μm above the 3 μm microparticle and 4 μm above the 9.75 μm microparticle. However, if we compare the light intensity at the focal plane of the 3 μm microparticle and the 9.75 μm microparticle, it should be noticed that the signal enhancement from the big microparticle is not as good as that from the small microparticle, a conclusion matching the experimental results.

The simulation results show that the spherical microparticle plays the role of a microlens focusing the radiation into extremely small regions and enhancing the signal, producing a "photonic nanojet". The latter is defined as the beam that emerges from a dielectric microsphere illuminated by a plane wave; it has a high intensity, subwavelength transverse dimensions, and low divergence.[40,41] The peak intensity of the optical field in the focusing zones can be several orders of magnitude larger than the intensity of the source, and the spatial dimension of the foci can be as small as fractions of the incident wavelength.[42] Hence, the use of photonic nanojets holds great potential for single-molecule or nanoparticle detection and for nanolithographic applications.[41,43,44] By properly selecting the optical properties of the particulate material and the microparticle size, it is possible to optimize the setup and find conditions that lead to a maximum intensity and minimum half-width of the nanojet. Figure 4a shows the

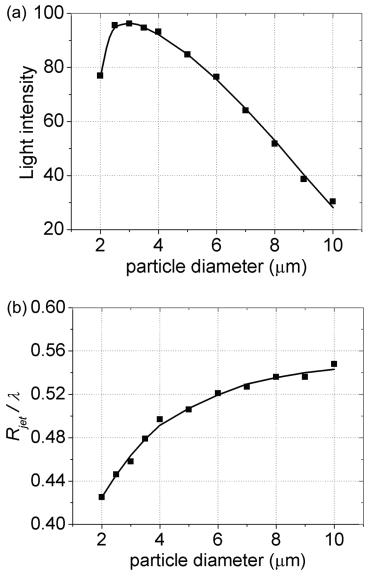

Figure 4. (a) Maximum intensity on the focal plane and (b) half-width of the photonic nanojet normalized by the wavelength of a melamine microparticle as function of the microparticle diameter. The dots are values obtained from the simulation, and the solid curves are guides to the eye.

calculated signal intensity on the focal plane, when using microparticles of different sizes. The maximum intensity is obtained with a 3 μm microparticle. In Figure 4b, R_{jet} is the half-width of the nanojet and λ = 668 nm is the wavelength. The larger the microparticle is, the wider the photonic nanojet, and associated with this, the intensity on the focal plane decreases significantly.

Moreover, the influence of the microparticle refractive index is studied. Their ratio n_p/n_m is defined as the optical contrast. Figure 5 shows the maximum intensity and normalized half-width of the photonic nanojet for a 3 μm microparticle as a

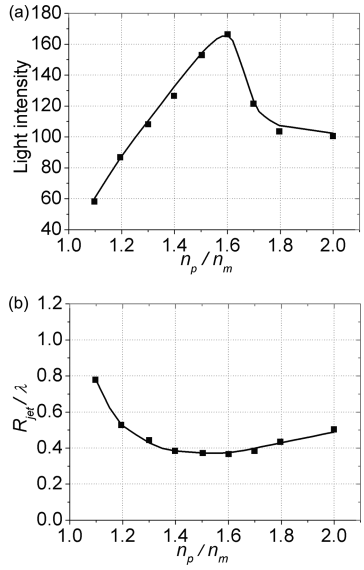

Figure 5. (a) Maximum intensity on the focal plane and (b) normalized half-width of the photonic nanojet of a melamine microparticle as the functions of optical contrast n_p/n_m.

function of the optical contrast, without taking into account light absorbance (κ = 0). The increase of the optical contrast leads to an increase in peak intensity of the photonic nanojet (see Figure 5a). We find that, as the optical contrast increases, the external focal point of the nanojet gets closer to the microparticle, and at a certain value of the optical contrast (n_p/n_m = 1.6), the center of the external focal waist coincides with the microparticle surface, which corresponds to the maximum of the intensity. This condition corresponds to a refractive index n_p = 2.13 in water. Increasing the optical contrast beyond 1.6, the external focal point shifts into the microparticle, and the photonic nanojet which "leaks out" of the microparticle shows lower intensity.

The classical diffraction limit of an objective or lens is defined as $\Delta x = \lambda/(2\mathrm{NA})$. For the objective used in the experiment (Zeiss Achroplan, 20×, NA = 0.45), the value of $\Delta x/\lambda$ is ~1.1. When this value is compared to the simulation shown in Figure 5b, the half-width of the photonic nanojet has subdiffraction limit values over a very wide range of the optical contrast. The size of the nanojet is correlated to the peak intensity, and at an optical contrast of 1.6, the smallest calculated nanojet size is ~0.18λ. When the numerical aperture of the microscope objective gets bigger, its associated diffraction limit will get smaller and will approach more the nanojet half-width. For example, when a water-immersion objective (NA = 1.20) would have been used, the optical contrast should be in the range of 1.34, 1.72 in order for the nanojet to be better focused than the resolution offered by the objective. When the optical contrast is 1.26 (n_p = 1.68, n_m = 1.33), the normalized half-width of the nanojet is 0.48. The "effective NA" of the microparticle can be calculated from the equation $\Delta x/\lambda = R_{jet}/\lambda$, i.e., 1/(2NA) = 0.48; hence, the "effective NA" of the 3 μm microparticle is 1.04. This calculation means that the use of a 3 μm microparticle would permit a resolution provided by an objective with NA = 1.04.

In an assay that relies on the measurement of fluorescent signals, background noise may be created by the instrumentation used to detect the fluorescence, autofluorescence of the substrate to which the molecules are linked (in case of a

heterogeneous assay), and signal due to the nonspecific binding of the labeled antibodies. While the fluorescence related to the two last noise sources and the "true" signal in the experiment are both collected by the microparticle and get enhanced with the same amplification factor, use of the microlens will not increase the signal-to-noise ratio. However, a microlens allows the maximum fluorescence intensity at its focal plane to be amplified compared to the intensity generated by the source, and evidently this implies a better signal-to-noise ratio, when considering instrumentation noise. We have also calculated the fluorescence enhancement when either a planar linear light source or a source with 3D topography is used. The simulation of Figure 6a corresponds to the case where a 3 μm microparticle is immobilized on a 3D APTES 3 μm dot microstructure and shows that the fluorescence on the microparticle's focal plane is amplified ~7-fold with respect to the center of the dot microstructure and ~3.6-fold compared to the intensity at the ridge of the microstructure. We note that this factor 7 is well above the experimentally determined enhancement factor of 2.8. The reduced experimental fluorescent signal is most likely related to the experimental observation conditions, in particular the use of a 550 μm thick glass substrate in combination with a 20× microscope objective with coverglass correction of only 0.17 (meaning this objective best performs using a substrate of up to 170 μm in thickness); spherical aberration is known to increase with thickness of the coverglass that is placed in the optical path between the front of the objective and the fluorescent source. From the simulation of Figure 6b, we find that, when a 3 μm microparticle would be immobilized on a planar APTES 3 μm dot microstructure (without the ridges), the fluorescence on the microparticle's focal plane is amplified ~3-fold.

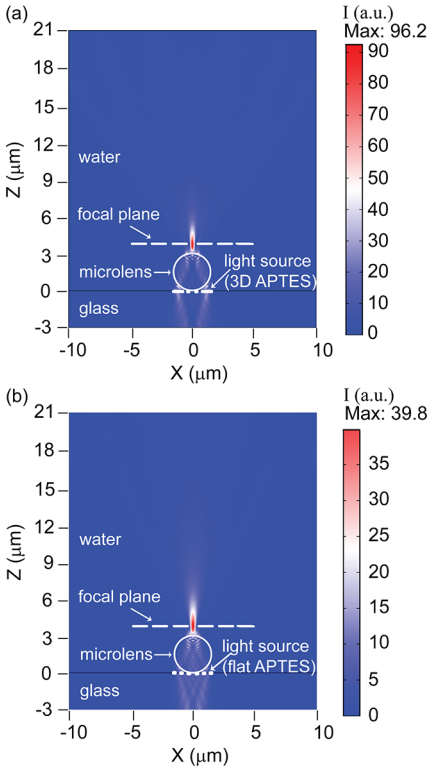

Figure 6. Simulation result of the fluorescence enhancement of a 3 μm melamine microlens, taking (a) a 3D and (b) a planar light source, corresponding to the use of a 3D APTES microstructure and a flat APTES layer, respectively.

Since the 20× objective (NA 0.45) used in the experiments enables imaging with a lateral resolution of 742 nm at the emission wavelength of the detection Abs, the fluorescent intensity generated by the immunoassay can be measured with sufficient spatial resolution from the individual dots and squares. The setup allows detecting in parallel many fluorescent sites, the number of which can be further increased via two approaches: (i) decreasing the size of an individual APTES pattern and (ii) increasing the density of the APTES microstructures. While in principle it is possible to realize APTES microstructures smaller than 3 μm to increase the number of reaction sites, microfabrication becomes challenging as one approaches the limit of classical lithography. Also, our numerical simulations (see Figure 4a) and the calculations shown in ref 42 suggest that optimum focusing of a microlens and maximal fluorescent intensity is obtained when using a microparticle of 3 μm. Keeping therefore this as minimum size for a fluorescent site, we estimate that the 3 μm dot APTES pattern in principle could be densified by a factor ~5 by limiting the space between the individual dots without compromising the assembly of the 3 μm microlenses. Therefore, our assay format in principle can be made compatible with that of digital ELISA,[45,46] which was reported for single-molecule analysis in femtoliter-sized wells (with a diameter of 4.5 μm, a pitch of ~10 μm, and depth of 3.25 μm). On the other hand, our present design still compares very favorably with common microarray spots (~100 μm diameter);[11] indeed, the 3 μm dots represent a reduction by a factor 30 in diameter and 900 in area, while the 8 μm squares represent a 12- and 144-fold reduction in diameter and area, respectively.

Also we think that our approach can be used for a multiplexed immunoassay. The principle of multiplex detection on magnetic beads placed at the bottom of a microfluidic channels was demonstrated before by Sivagnanam et al.[26] For incubation with a solution carrying different Ags, different capture Abs can be immobilized on the micropatterns. Thereafter, the target Ags can be bound with their corresponding capture Abs with high specificity and then distinguished by the detection Abs labeled by different fluorophores.

■ CONCLUSIONS

We have patterned 3D APTES microstructures on a glass wafer, in particular microtextured dots and squares for enhanced (2–3×) immunofluorescent detection with respect to the signal originating from a planar substrate. In situ positioning microlenses on top of the fluorescent APTES microstructures allowed concentration of the fluorescence intensity by an additional factor of 2.8 when using a microlens of 3 μm. FEM simulations corresponding to our experimental conditions showed the photonic nanojet that emerged from a microparticle and verified the enhancement of the fluorescent signal, its dependence on microparticle size, on the optical contrast, and the use (or not) of a 3D microstructured light source. We think that our approach offers high potential and can play a role in lowering fluorescent detection limits. We also mention that an important advantage of using a microlens is that the local enhancement of fluorescence can be simply monitored by using a low magnification/low numerical aperture microscope objective. Focusing of such objective is easy, the associated

large field-of-view allows many light sources to be observed at once, and a statistical study of the intensity distribution over the different spots is possible. We anticipate that the reported method can be easily adapted to other types of fluorescent immunoassays and further used in multiplex detection formats.

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Notes

The authors declare no competing financial interest.

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